
1. 【简答题】

For a carrier wave, $f_c=100$ MHz, $U_{cm}=5$ V. The modulation signal has an expression, $u_{\Omega}(t)=\cos 2\pi \times 10^3 t + 2\cos 2\pi \times 500 t$. Please write out the expression of the FM wave, assuming that both modulation signal has the same $\Delta f_{\max}=20$ KHz.

答案:

$$u_{FM}(t) = U_{cm} \times \cos(2\pi \times f_c t + m_{f1} \sin \Omega_1 t + m_{f2} \sin \Omega_2 t)$$

$$\text{Since } \Delta f_{\max 1} = \Delta f_{\max 2} = \Delta f_{\max} = m_f \Omega / 2\pi = 20 \text{ KHz}$$

$$\therefore m_{f1} = (2\pi \times \Delta f_{\max}) / \Omega_1 = 2\pi \times 20 \times 10^3 / 2\pi \times 10^3 = 20,$$

$$m_{f2} = (2\pi \times \Delta f_{\max}) / \Omega_2 = 2\pi \times 20 \times 10^3 / 2\pi \times 500 = 40$$

$$\therefore u_{FM}(t) = U_{cm} \times \cos(2\pi \times 10^8 t + m_{f1} \sin \Omega_1 t + m_{f2} \sin \Omega_2 t) = 5 \cos(2\pi \times 10^8 t + 20 \sin 2\pi \times 10^3 t + 40 \sin 2\pi \times 500 t)$$

2. 【简答题】

An AM and FM wave has the same carrier wave frequency and modulation signal. $f_c=1$ MHz. $u_{\Omega}(t)=0.1 \sin 2\pi \times 10^3 t$. It is known that, for the frequency modulation, the frequency shift caused by unit modulation voltage is 1 KHz/V.

- (1) Please calculate B_{AM} (the bandwidth of the AM wave) and B_{FM} (the effective bandwidth of the FM wave).
- (2) When the modulation signal changed into $u_{\Omega}(t)=20 \sin 2\pi \times 10^3 t$, please recalculate B_{AM} and B_{FM} .

答案:

$$(1) \text{ for FM, } \Delta f_{\max} = K_f U_{\Omega m} / 2\pi \therefore K_f = 2\pi \times 1 \text{ KHz/V}$$

$$\therefore m_f = (K_f U_{\Omega m}) / \Omega = 2\pi \times 10^3 \times 0.1 / 2\pi \times 10^3 = 0.1$$

$$\therefore u_{FM}(t) = U_{cm} \times \cos(2\pi \times f_c t + m_f \sin \Omega t) = U_{cm} \times \cos(2\pi \times 10^6 t + 0.1 \sin 2\pi \times 10^3 t)$$

$$B_{FM} = 2(m_f + 1)F = 2.2 \times 10^3 \text{ Hz}$$

$$B_{AM} = 2F = 2 \times F = 2 \times 10^3 \text{ Hz}$$

$$(2) B_{AM} = 2F = 2 \times F = 2 \times 10^3 \text{ Hz}$$

$$\text{When } u_{\Omega}(t) = 20 \sin 2\pi \times 10^3 t, m_f = (K_f U_{\Omega m}) / \Omega = 2\pi \times 10^3 \times 20 / 2\pi \times 10^3 = 20$$

$$B_{FM} = 2(m_f + 1)F = 42 \times 10^3 \text{ Hz}$$

3. 【简答题】

For a modulation signal, $F=400$ Hz, $U_{\Omega m}=2.4$ V, the modulation factor is 60. When the modulation signal's frequency reduced to 250 Hz and the amplitude increased to 3.2 V, what value would the modulation factor changed into?

答案:

(1) for FM,

$$m_f = (K_f U_{\Omega m}) / \Omega = (K_f U_{\Omega m}) / 2\pi F$$

$$\therefore K_f \times 2.4 / 2\pi \times 400 = 60$$

$$\therefore K_f = 60 \times 2\pi \times 400 / 2.4$$

When $U_{\Omega m}$ changed into 3.2 V, F changed into 250 Hz

$$m_f = (K_f U_{\Omega m}) / 2\pi F = (60 \times 2\pi \times 400 / 2.4) \times 3.2 / 2\pi \times 250 = 128$$

(2) for the PM,

$$m_p = K_p U_{\Omega m}$$

$$\therefore K_p = U_{\Omega m} / m_p = 2.4 / 60$$

When $U_{\Omega m}$ changed into 3.2 V, F changed into 250 Hz

$$m_p = K_p U_{\Omega m} = (2.4 / 60) \times 3.2 = 80$$

4. 【简答题】

The frequency discrimination characteristics curve has been shown in Fig. 1.

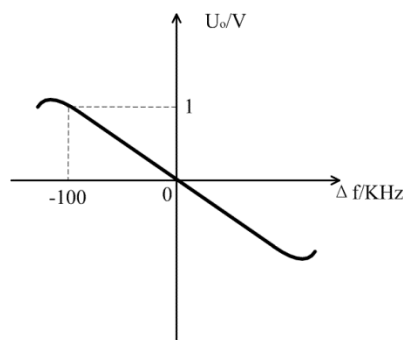
Known that the output voltage of the frequency discriminator is

$$u_o(t) = \cos 4\pi \times 10^3 t \text{ (V)}$$

Please calculate:

(1) Frequency discriminating trans-conductance g_d .

(2) The mathematical expression of the input signal $u_{FM}(t)$ and the modulation signal $u_{\Omega}(t)$



Ex.Fig.5.1

答案:

$$g_d = \Delta u / \Delta f = -1/100 \text{ (V/KHz)}$$

$$\text{Suppose the } u_{FM}(t) = U_{cm} \times \cos(2\pi \times f_c t + m_f \sin 2\pi F t)$$

$$\text{For } u_o(t) = \cos 4\pi \times 10^3 t \text{ (V)}, \Delta u_{max} = 1 \text{ V}, \therefore \Delta f_{max} = \Delta u_{max} / g_d = -100 \text{ KHz}$$

$$\text{Because } \Delta f_{max} = K_f U_{\Omega m} / 2\pi = m_f \times F$$

$$\therefore m_f = \Delta f_{max} / F = 100 \times 10^3 / 2 \times 10^3 = -50$$

$$\therefore u_{FM}(t) = U_{cm} \times \cos(2\pi \times f_c t - 50 \sin 4\pi \times 10^3 t)$$

$$u_{\Omega}(t) = -U_{\Omega m} \times \cos(4\pi \times 10^3 t)$$